

COURSE NAME : TOOL ENGINEERING DRAWING -1
COURSE CODE : 3112
COURSE CATEGORY : B
PERIODS / WEEK : 6
PERIODS/SEMESTER : 90
CREDITS : 6

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Basic Fastening Devices, Rivetted joints Limits, Fits and Tolerance, Surface finish	30
2	Assembly and Detailed drawing	20
3	Jigs	20
4	Fixtures	20
Total		90

COURSE OUTCOMES

After completion of the study the students will be able to:

1. Know the need of fastening device.
2. Understand the significance of limit, fit, tolerance.
3. Make the detailed drawing of given assembly drawings.
4. Understand jigs and fixtures.

SPECIFIC OUTCOMES

MODULE– I

1.1.0 Know the need for fastening devices.

- 1.1.1 Identify the basic fastening devices.
- 1.1.2 Draw the standard thread profiles.
- 1.1.3 Draw Hexagonal headed bolt, nut and bolted joints.
- 1.1.4 Draw joints using allenscrew, grubscrew.
- 1.1.5 Draw locking nuts.

- 1.1.6 Understand various types of rivet heads.
- 1.1.7 Draw riveted joints
- 1.1.8 Calculate the pitch and arrangements of rivet in rows (Lap and But joints).

1.2.0 Understand the significance of Limit, Fit, Tolerance.

- 1.2.1 Understand the role of dimension to obtain a fit as per BIS.
- 1.2.2 Define limits, fits & tolerances.

1.3.0 Select dimension from standards to give different types of fit for a given mating part Simple problems with examples . Familiarization of tables

- 1.3.1 Compute fit and tolerance from tables.
- 1.3.2 Understand the need of surface finish.
- 1.3.3 Identify the standard symbols to represent surface finishing.

MODULE II

2.1.0 Apply assembly and detailed drawing.

- 2.1.1 Select the appropriate size of the sheet for preparing a required assembly/detailed drawing
- 2.1.2 List the sequence of steps for preparing assembly/detailed drawing.
- 2.1.3 Create assembly drawings with the drawings of components such as tool maker's clamps, and equalizing clamps.
- 2.1.4 Create detailed drawings given the assembly drawings such as clamps, couplings, turn buckle and equalizing clamp
- 2.1.5 Construct the parts list of assembly drawing.
- 2.1.6 Understand the conventional representation of machine parts.

MODULE III

3.1.0 Recognize the various elements of Jigs

- 3.2.1 Distinguish between various types of bushes , locating pins and clamping devices.
- 3.2.2 Draw the detailed views of jigs like plate jig, box jig and index jig.
- 3.2.3 Draw each parts drawing according to production manner and prepare bill of material of above Jigs. Compute fit and tolerance from tables.

MODULE IV

4.1.0 Identify the various elements of Fixture and its principles

- 4.1.1 Draw the assembled and dismantled drawings of Fixtures for turning, milling, grinding.

CONTENT OUTLINES

MODULE I

BASIC FASTENING DEVICES

Temporary and permanent fastening devices – standard thread profiles – areas of application of Bolts, Nuts, Screws (Allen Screw and Grub Screw) – different types of Bolted joints and screw joint for different applications – purpose of locking nuts – types of lock nuts – proportions and specifications of rivet – riveted joints (Lap and Butt) – calculations of pitch and arrangement of rivets in rows and the joints using Standard Proportions – Selection of appropriate fastening devices to meet the given working conditions Draw lap joint– Butt joint (single row, double row)– Single cover – Double cover – Calculation of pitch and arrangement of rivets – specification of rivets.

LIMITS, FITS & TOLERANCE

Definition of limits, fits & tolerances – select dimensions from BIS to obtain clearance, transition and interference fits for a given set of mating parts. Computation of fit and tolerances from the table .preparation of drawings of mating parts and representation of fits and tolerances – representation of surface finish and need on the drawings as per BIS.

Exercise in computing tolerances - simple problems

Examples for hole basis systems, shaft basis systems.

MODULE– II

ASSEMBLY AND DETAILED DRAWING

Needs and functions of assembly and detailed drawings – selection of sheet size – preparation of title block – bill of materials– steps in preparing assembly and detailed drawing –

Exercise in preparing assembly and detailed drawing of commonly available components such as clamps, flanged couplings- turnbuckle and equalizing clamps.

Conventional representation of machine parts- screw thread –internal and external thread –ratchet and pinion – repeated parts.

MODULE III

JIGS

Components of jigs and fixtures – different types of bushes – locating pins and clamping devices – types of jigs – plate jig, box jig, indexing jig drawings needed for the purpose of production.

Preparation of bill of materials.

MODULE IV

FIXTURES

Prepare relevant views of the parts of a given fixtures for the purpose of production. Discussion the views with relevant notes and indications –Fixtures- turning fixture, milling fixtures, grinding fixture, detailed dimensioning, and preparation of bill of material.

REFERENCE BOOKS

1. Jigs & Fixtures - Joshy
2. Production Drawing - John & Varghese
3. Machine Drawing - N.D. Bhat
4. Design Data Hand Book - PSG
5. Jigs & Fixtures - Grant
6. Machine Drawing - N. D Junnarkar